RACE
Regular Analysis Through Continuous Exercise

Methods of Differentiation : RACE-01

Single Correct Question

1. If $y = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots \infty$, then $\frac{dy}{dx} =$

(A) y (B) $y - 1$ (C) $y + 1$ (D) None of these

MDF153

2. $\frac{d}{dx}\left(e^x \log \sin 2x\right) =$

(A) e^x (log sin $2x$ + 2 cot $2x$)

(B) e^x (log cos $2x$ + 2 cot $2x$)

(C) e^x (log cos $2x$ + cot $2x$)

(D) None of these

MDF154

3. $\frac{d}{dx}\left[\left(\frac{\tan^2 2x - \tan^2 x}{1 - \tan^2 2x \tan^2 x}\right)\cot 3x\right]$

(A) tan $2x$ tan x (B) tan $3x$ tan x (C) sec^2 x (D) sec$x$ tan$x$

MDF155

4. If $f(x) = |\cos x|$, then $f'\left(\frac{3\pi}{4}\right)$ is equal to -

(A) $-\frac{1}{\sqrt{2}}$ (B) $\frac{1}{\sqrt{2}}$ (C) 1 (D) –1

MDF156

Subjective Type Question

5. Differentiate each of following functions by first principle :

(i) $f(x) = \tan x$ (ii) $f(x) = e^{\sin x}$

MDF157

Differentiate the given functions with respect to x :

6. $y = \sqrt{1 + \tan\left(x + \frac{1}{x}\right)}$

MDF158

7. $y = x^2 \log_3 x$

MDF159

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8. $y = \log_2[\log_3(\log_5 x)]$

MDF160

9. $y = Ae^{-k^2x}\sin(\omega x + \alpha)$

MDF161

10. If $y = e^x \tan x + x \log_e x$, find $\frac{dy}{dx}$.

MDF162

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ANSWER KEY

Question	1	2	3	4	
Answer	A	A	C	B	

6. $\frac{x^2-1}{2x^2\cos^2\left(x+\frac{1}{x}\right)\sqrt{1+\tan\left(x+\frac{1}{x}\right)}}$

7. $2x\log_3 x + \frac{x}{\ell n 3}$

8. $\frac{1}{x\log_5 x\log_3(\log_5 x)\ell n 2\ell n 3\ell n 5}$

9. $Ae^{-k^2x}[\omega\cos(\omega x+\alpha) - k^2\sin(\omega x+\alpha)]$

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RACE
Regular Analysis Through Continuous Exercise

Methods of Differentiation : RACE-02

Single Correct Question

1. If $y = e^{x+e^{x+e^{x+\ldots\infty}}}$, then $\frac{dy}{dx}$ =

(A) $\frac{y}{1-y}$ (B) $\frac{1}{1-y}$ (C) $\frac{y}{1+y}$ (D) $\frac{y}{y-1}$

MDF163

2. $\frac{d}{dx}\{(\sin x)^x\}$ =

(A) $\left[\frac{x\cos x+\sin x\log\sin x}{\sin x}\right]$ (B) $(\sin x)^x\left[\frac{x\cos x+\sin x\log\sin x}{\sin x}\right]$

(C) $(\sin x)^x\left[\frac{x\sin x+\sin x\log\sin x}{\sin x}\right]$ (D) None of these

MDF164

3. If $x = \frac{1-t^2}{1+t^2}$ and $y = \frac{2t}{1+t^2}$, then $\frac{dy}{dx}$ =

(A) $\frac{-y}{x}$ (B) $\frac{y}{x}$ (C) $\frac{-x}{y}$ (D) $\frac{x}{y}$

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MDF165

4. If $y=\frac{1}{4}u^4, u=\frac{2}{3}x^3+5$, then $\frac{dy}{dx}$ =

(A) $\frac{1}{27}x^2(2x^3+15)^3$ (B) $\frac{2}{27}x(2x^3+5)^3$

(C) $\frac{2}{27}x^2(2x^3+15)^3$ (D) None of these

MDF166

5. If $x = a(t - \sin t), y = a(1 + \cos t)$, then $\frac{dy}{dx}$ equals-

(A) $-\tan\frac{t}{2}$ (B) $\cot\frac{t}{2}$ (C) $-\cot\frac{t}{2}$ (D) $\tan\frac{t}{2}$

MDF167

Integer Type Question

6. Let $f(x) = (x-1)^{100}\ (x-2)^{2(99)}\ (x-3)^{3(98)} \ldots (x-100)^{100}$, then the value of $\frac{f'(101)}{f(101)}$ is

MDF168

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Subjective Type Question

Differentiate the given functions with respect to x :

7. $y=\dfrac{(x+1)^3\sqrt[4]{x-2}}{\sqrt[5]{(x-3)^2}}$

MDF169

8. $y=(x^2+1)^{\sin x}$

MDF170

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ANSWER KEY

Question	1	2	3	4	5	6
Answer	A	B	C	C	C	5050

7. $\dfrac{57x^2-302x+361}{20(x-2)(x-3)}.\dfrac{(x+1)^2\sqrt[4]{x-2}}{\sqrt[5]{(x-3)^2}}$ **8.** $(x^2+1)^{\sin x}\left[\dfrac{2x\sin x}{x^2+1}+\cos x\ell n\left(x^2+1\right)\right]$

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RACE
Regular Analysis Through Continuous Exercise

Methods of Differentiation : RACE-03

Single Correct Question

1. If $x\sqrt{1+y}+y\sqrt{1+x}=0$, then $\frac{dy}{dx}$ =

(A) $1+x$ (B) $(1+x)^{-2}$ (C) $-(1+x)^{-1}$ (D) $-(1+x)^{-2}$

MDF171

2. If $2^x+2^y=2^{x+y}$, then $\frac{dy}{dx}$ is equal to –

(A) $\frac{2^x+2^y}{2^x-2^y}$ (B) $\frac{2^x+2^y}{1+2^{x+y}}$ (C) $2^{x-y}\left(\frac{2^y-1}{1-2^x}\right)$ (D) $\frac{2^x+y-2^x}{2^y}$

MDF172

3. If $y=\sqrt{\log x+\sqrt{\log x+\sqrt{\log x+...\infty}}}$, then $\frac{dy}{dx}$ =

(A) $\frac{x}{2y-1}$ (B) $\frac{x}{2y+1}$ (C) $\frac{1}{x(2y-1)}$ (D) $\frac{1}{x(1-2y)}$

MDF173

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4. If $x^3-y^3+3xy^2-3x^2y+1=0$, then at $(0,1)$ $\frac{dy}{dx}$ equals-

(A) 1 (B) –1 (C) 2 (D) 0

MDF174

5. If g is inverse of f and $f(x)=2x+\sin x$; then $g'(x)$ equals :

(A) $-\frac{3}{x^2}+\frac{1}{\sqrt{1-x^2}}$ (B) $2+\sin^{-1}x$ (C) $2+\cos g(x)$ (D) $\frac{1}{2+\cos(g(x))}$

MDF175

6. If $y\sec x+\tan x+x^2y=0$, then $\frac{dy}{dx}$ =

(A) $\frac{2xy+\sec^2 x+y\sec x\tan x}{x^2+\sec x}$ (B) $-\frac{\left(2xy+\sec^2 x+\sec x\tan x\right)}{x^2+\sec x}$

(C) $-\frac{\left(2xy+\sec^2 x+y\sec x\tan x\right)}{x^2+\sec x}$ (D) None of these

MDF176

7. If g is the inverse of f and $f'(x)=\frac{1}{1+x^3}$ then $g'(x)$ is equal to -

(A) $1+[g(x)]^3$ (B) $\frac{-1}{2(1+x^2)}$ (C) $\frac{1}{2(1+x^2)}$ (D) $\frac{1}{1+\left[g(x)\right]^3}$

MDF177

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Integer Type Question

8. Let $g(x) = f^{-1}(x)$ where $f(x) = x + e^x$ then find $2g'(1)$.

MDF178

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	
Answer	D	C	C	A	D	D	A	1	

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RACE
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Methods of Differentiation : RACE-04

Single Correct Question

1. If $y = (A + Bx)e^{mx} + (m-1)^{-2}\, e^{x}$ then $\frac{d^2y}{dx^2} - 2m\frac{dy}{dx} + m^2y$ is equal to -

(A) e^{x} (B) e^{mx} (C) e^{-mx} (D) $e^{(1-m)x}$

MDF179

2. $\frac{d}{dx}\left(\tan^{-1}\sqrt{\frac{1+\cos\frac{x}{2}}{1-\cos\frac{x}{2}}}\right)$, $x \in \left(\frac{\pi}{2}, \pi\right)$ is equal to

(A) $-\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $-\frac{1}{2}$ (D) $\frac{1}{4}$

MDF180

3. If $y = \tan^{-1}\frac{4x}{1+5x^2} + \tan^{-1}\frac{2+3x}{3-2x}$, $x > 0$, $x \neq \frac{3}{2}$ then $\frac{dy}{dx}$ =

(A) $\frac{1}{1+25x^2} + \frac{2}{1+x^2}$ (B) $\frac{5}{1+25x^2} + \frac{2}{1+x^2}$ (C) $\frac{5}{1+25x^2}$ (D) $\frac{1}{1+25x^2}$

MDF181

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4. If $y = \tan^{-1}\left[\frac{\sin x + \cos x}{\cos x - \sin x}\right]$, $x \in \left(0, \frac{\pi}{4}\right)$ then $\frac{dy}{dx}$ is

(A) $\frac{1}{2}$ (B) $\frac{\pi}{4}$ (C) 0 (D) 1

MDF182

Subjective Type Question

5. If $y = xe^{x^2}$ then find y''.

MDF183

6. Find y'' at $x = \pi/4$, if $y = x \tan x$.

MDF184

7. If $f(x) = \cos^{-1}(4x^3 - 3x)$, then find :

(a) $f'\left(-\frac{\sqrt{3}}{2}\right)$, (b) $f'(0)$, (c) $f'\left(\frac{\sqrt{3}}{2}\right)$

MDF185

8. Prove that the function $y = e^{x} \sin x$ satisfies the relationship $y'' - 2y' + 2y = 0$.

MDF186

ANSWER KEY

Question	1	2	3	4
Answer	A	A	C	D

5. $(y'' = 4y + 2xy')$ **6.** $(\pi + 4)$ **7.** (a) – 6 (b) 3 (c) –6

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RACE Regular Analysis Through Continuous Exercise

Methods of Differentiation : RACE-05

Single Correct Question

1. If f is differentiable in (0, 6) & $f'(4) = 5$, then $\underset{x\to 2}{\text{Limit}}\dfrac{f(4)-f(x^2)}{2-x}$ =

(A) 5 (B) 5/4 (C) 10 (D) 20

MDF187

2. Let $f(x)$ be differentiable at $x = h$, then $\underset{x\to h}{\text{Lim}}\dfrac{(x+h)f(x)-2hf(h)}{x-h}$ is equal to -

(A) $f(h) + 2hf'(h)$ (B) $2f(h) + hf'(h)$ (C) $hf(h) + 2f'(h)$ (D) $hf(h) - 2f'(h)$

MDF188

3. If $f_n(x)$, $g_n(x)$, $h_n(x)$, $n = 1, 2, 3$ are polynomials in x such that $f_n(a) = g_n(a) = h_n(a)$, $n = 1, 2, 3$ and $F(x)=\begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ g_1(x) & g_2(x) & g_3(x) \\ h_1(x) & h_2(x) & h_3(x) \end{vmatrix}$. Then $F'(a)$ is equal to

(A) 0 (B) $f_1(a)\, g_2(a)\, h_3(a)$ (C) 1 (D) None of these

MDF189

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4. $f(x)=\begin{vmatrix} x^3 & x^2 & 3x^2 \\ 1 & -6 & 4 \\ p & p^2 & p^3 \end{vmatrix}$, where p is a constant, then $\dfrac{d^3 f(x)}{dx^3}$ is

(A) Proportional to x^2 (B) Proportional to x (C) Proportional to x^3 (D) A constant

MDF190

5. If $y = \sin px$ and y_n is the nth derivative of y, then $\begin{vmatrix} y & y_1 & y_2 \\ y_3 & y_4 & y_5 \\ y_6 & y_7 & y_8 \end{vmatrix}$ is equal to

(A) 1 (B) 0 (C) –1 (D) None of these

MDF191

Integer Type Question

6. Evaluate the following limits using L'Hospital's Rule or otherwise :

$\underset{x\to 0}{\text{Lim}}\dfrac{x^{6000}-(\sin x)^{6000}}{x^2.(\sin x)^{6000}}$

MDF192

7. If $\underset{x\to 0}{Lim}\dfrac{1-\cos x.\cos 2x.\cos 3x.....\cos nx}{x^2}$ has the value equal to 253, find the value of n (where $n \in N$).

MDF193

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8. If $f(x)=\begin{vmatrix} 2x & x^2 & x^3 \\ x^2+2x & 1 & 3x+1 \\ 2x & 1-3x^2 & 5x \end{vmatrix}$, then find $f'(1)$.

MDF194

Subjective Type Question

9. Using L'Hospital rule find

(a) $\lim\limits_{x\to 0}\dfrac{\tan x - x}{x^3}$ (b) $\lim\limits_{x\to 0}\dfrac{e^x - x - 1}{x^2}$

MDF195

10. Using L'Hospital rule verify that :

(a) $\lim\limits_{x\to 0}\dfrac{\sin x - \tan x}{x^3} = -\dfrac{1}{2}$ (b) $\lim\limits_{x\to 0}\dfrac{\ell n(1+x)}{x} = 1$

MDF196

11. Given a real valued function $f(x)$ as follows :

$f(x) = \dfrac{x^2+2\cos x-2}{x^4}$ for $x < 0$; $f(0) = \dfrac{1}{12}$ & $f(x) = \dfrac{\sin x - \ln(e^x \cos x)}{6x^2}$ for $x > 0$. Test the continuity and differentiability of $f(x)$ at $x = 0$.

MDF197

12. If $f(x)=\begin{vmatrix} e^x & x^2 \\ \ell nx & \sin x \end{vmatrix}$, then find $f'(1)$.

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MDF198

ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	D	A	A	D	B	1000	11	9

9. **(a)** $\dfrac{1}{3}$ **(b)** $\dfrac{1}{2}$

11. f is cont. but not derivable at $x = 0$.

12. $e(\sin 1 + \cos 1) - 1$

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